Robust Galaxy Morphology Classification with Galaxy Zoo 2 Under Observational Domain Shift

Danyal Ahmed

March 2026

Abstract

This project aims to study automated galaxy morphology classification using Galaxy Zoo 2 (GZ2) under realistic observational domain shift. Using the debiased vote fractions derived by Hart et al.[1], I plan to construct a 4-class taxonomy derived from the GZ2 decision tree (Figure A.1): (1) smooth/elliptical, (2) edge-on disk, (3) face-on spiral, and (4) face-on non-spiral or lenticular. I will compare three model families: (i) regularized multinomial logistic regression (or linear SVM), (ii) a tree ensemble baseline (random forest/gradient boosting), and (iii) a convolutional neural network (CNN) in PyTorch trained on whole-image pixel inputs. Results will be reported using macro F1 scores, confusion matrices, and performance-versus-redshift curves to quantify robustness and characterize failure modes, with the goal of identifying which modeling approaches generalize best as image quality degrades.

1 Introduction

Large-scale astronomical imaging surveys contain millions of galaxies, but much of the scientific analysis coming from these surveys require morphological labels. These labels are traditionally obtained through expert or citizen-science inspection, which is time-consuming and cannot scale to the volume of modern surveys. The goal is to develop and better understand automated methods for galaxy morphology classification from optical images, with an emphasis on robustness to the degradation of observational conditions.

1.1 Problem Description

The target is a supervised image-classification model that utilizes the Galaxy Zoo 2 (GZ2) dataset.[1] Each sample consists of an SDSS galaxy image x and a GZ2 vote-fraction annotation. I define a tree-consistent four-class morphology task aligned with the GZ2 decision tree (Figure A.1): (1) smooth/elliptical, (2) edge-on disk, (3) face-on spiral, and (4) face-on non-spiral or lenticular.

After converting vote fractions into hard class labels using fixed thresholds and discarding ambiguous low-confidence cases, the primary learning problem is to learn a classifier that maps an input image to one of these four classes.

1.2 Motivation

These labels are fundamental descriptors in studies of galaxy formation and evolution, correlating with stellar populations and star-formation rate, and allowing us to map cosmic structures such as galactic filaments and the universe's large-scale structure (LSS). Given the immense number of galaxies visible to us now, the development of automated morphology classification is likely a fundamental component for the future of astronomy and astrophysics.

2 Proposed Project

2.1 Features and preprocessing

Pixel input: Original GZ2 images are typically 424×424. I will resize images to $H \times W \times 3$ (planned: $H = W =$[128 or 224]) and normalize them.
Classical features: for non-DNN baselines I will extract an engineered morphology vector $\phi(x) \in \mathbb{R}^p$ (target $p \approx$ 50–200) consisting of radial intensity profile statistics, low-order shape moments, and a compact gradient/texture histogram; These will be standardized using training-set statistics.
Labels: GZ2 contains 239,695 samples paired with Hart16 debiased vote-fractions. For each sample image x_i I will derive a 4-class, tree-consistent label $y_i \in \{1, 2, 3, 4\}$ using GZ2 tasks t00 (smooth vs. features/disk), t01 (edge-on), and t03 (spiral): *smooth*, *edge-on disk*, *face-on spiral disk*, and *face-on non-spiral disk*. Vote fractions will be thresholded to retain high-confidence labels; ambiguous cases are discarded.

I will use three GZ2 data products joined by SDSS DR7 object ID (`objid`):

- **Hart16 Table 1 (debiased vote fractions):** label source for t01/t02/t04. [1]
- **GZ2 SDSS metadata (`gz2sample`):** redshift/photometry/size covariates used to define observational domains.[2]
- **Zenodo GZ2 images + mapping:** image cutouts and `objid`→`asset_id` mapping to locate each image.[2]

I plan to adjust the threshold such that I maintain an absolute minimum of 20,000 samples, 2,000 samples per-class, and 200 out-of-domain cases per-class. Given the immense number of samples, I hope to maintain a total sample space $\geq 100,000$.
Metadata: I define observational domains using `gz2sample` covariates (redshift, brightness, and apparent size). I will evaluate both IID splits and OOD splits that train on an "easy" regime (low-z/bright/large) and test on a "hard" regime (high-z/faint/small), with hyperparameters tuned only on the training domain. These may also be valuable as secondary features, concatenated with computed/learned feature embeddings.

2.2 Planned models and evaluation

I plan to compare three model families: (i) regularized multinomial logistic regression (or linear SVM) on $\phi(x)$, (ii) a tree ensemble (random forest or gradient boosting) on $\phi(x)$, and (iii) a CNN on pixels. I will report macro-F1, confusion matrices, and performance-versus-redshift curves, including IID→OOD degradation.

References

[1] Ross E. Hart, Steven P. Bamford, Kyle W. Willett, Karen L. Masters, Carolin Cardamone, Chris J. Lintott, Robert J. Mackay, Robert C. Nichol, Christopher K. Rosslowe, Brooke D. Simmons, and Rebecca J. Smethurst. Galaxy zoo: comparing the demographics of spiral arm number and a new method for correcting redshift bias. *Monthly Notices of the Royal Astronomical Society*, 461(4):3663–3682, 07 2016.

[2] Kyle W. Willett, Chris J. Lintott, Steven P. Bamford, Karen L. Masters, Brooke D. Simmons, Kevin R. V. Casteels, Edward M. Edmondson, Lucy F. Fortson, Sugata Kaviraj, William C. Keel, Thomas Melvin, Robert C. Nichol, M. Jordan Raddick, Kevin Schawinski, Robert J. Simpson, Ramin A. Skibba, Arfon M. Smith, and Daniel Thomas. Galaxy zoo 2: detailed morphological classifications for 304 122 galaxies from the sloan digital sky survey. *Monthly Notices of the Royal Astronomical Society*, 435(4):2835–2860, 09 2013.

A Figures

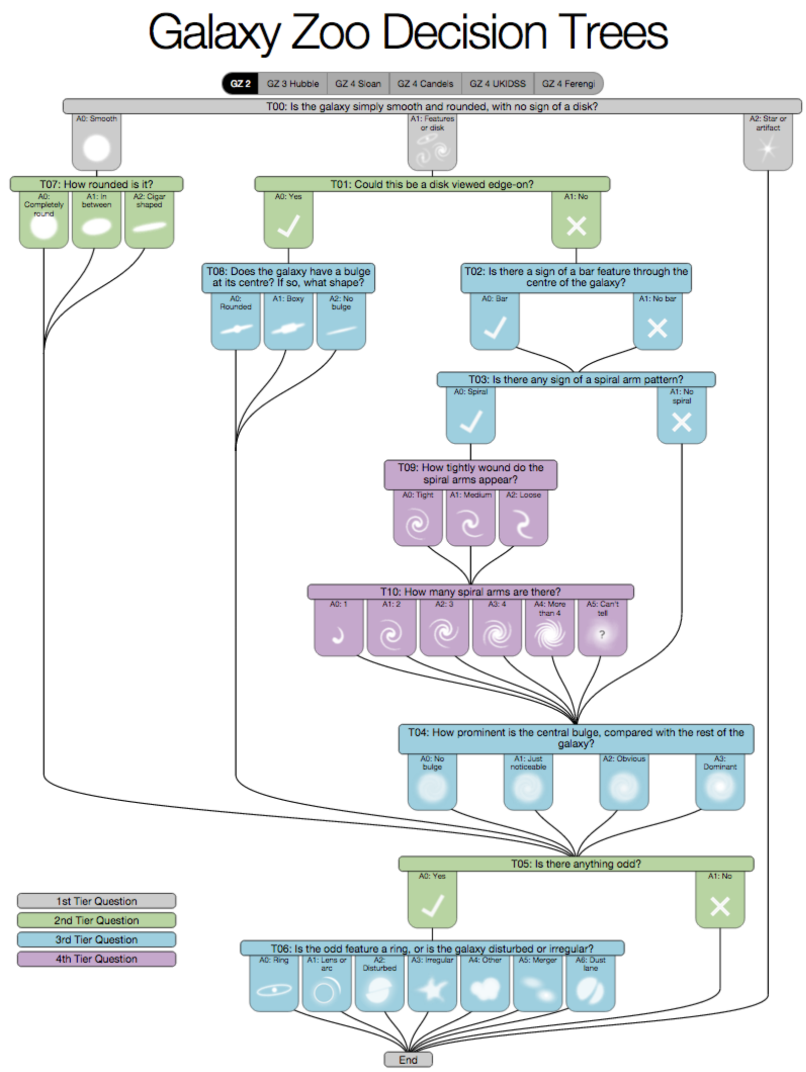

Figure A.1: Galaxy Zoo 2 Decision Tree